

Vector Calculus

Unit III: Vector Differentiation · Unit IV: Vector Integration
Lecture Notes

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Unit III: Vector Differentiation

1 The Vector Differential Operator

In vector calculus, any point $P(x, y, z)$ in space can be represented by the position vector

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}, \quad r^2 = |\vec{r}|^2 = x^2 + y^2 + z^2.$$

If $\phi(x, y, z)$ is a scalar function representing some quantity at any point $P(x, y, z)$, then the total change in ϕ is given by

$$\begin{aligned} d\phi &= \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = \left(\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z} \right) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k}) \\ &= \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z} \right) \phi \cdot d\vec{r}, \end{aligned}$$

$$d\phi = \nabla\phi \cdot d\vec{r}, \quad \text{where} \quad \nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}.$$

The operator ∇ (read “del” or “nabla”) has different meanings depending on the function it operates on:

- $\nabla\phi$ is known as the *gradient* of ϕ (ϕ a scalar function);
- $\nabla \cdot \vec{f}$ is known as the *divergence* of $\vec{f}$ ($\vec{f}$ a vector-valued function);
- $\nabla \times \vec{f}$ is known as the *curl* of $\vec{f}$ ($\vec{f}$ a vector-valued function).

2 Gradient, Directional Derivative and Normals

Note. • The directional derivative of ϕ is maximum in the direction of $\nabla\phi$, and its maximum value is $|\nabla\phi|$.

- $\nabla\phi$ is a normal vector to the surface $\phi(x, y, z) = c$; hence the unit normal is $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$.
- The angle θ between two surfaces at a common point is the angle between their normals:

$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1||\nabla\phi_2|}.$$

2.1 Problems

Problem 2.1. In what direction from the point $(1, 3, 2)$ is the directional derivative of $\phi = 2xz - y^2$ maximum? Also find its maximum value.

Solution. We know that the directional derivative of ϕ is maximum in the direction of $\nabla\phi$.

$$\begin{aligned} \nabla\phi &= \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z} \right) (2xz - y^2) = \vec{i}\frac{\partial}{\partial x}(2xz - y^2) + \vec{j}\frac{\partial}{\partial y}(2xz - y^2) + \vec{k}\frac{\partial}{\partial z}(2xz - y^2) \\ &= \vec{i}(2z) + \vec{j}(-2y) + \vec{k}(2x) = 2z\vec{i} - 2y\vec{j} + 2x\vec{k}. \end{aligned}$$

$$\nabla\phi(1, 3, 2) = 4\vec{i} - 6\vec{j} + 2\vec{k}.$$

The directional derivative is maximum in the direction $4\vec{i} - 6\vec{j} + 2\vec{k}$, and the maximum magnitude is

$$|\nabla\phi| = \sqrt{4^2 + (-6)^2 + 2^2} = \sqrt{16 + 36 + 4} = \sqrt{56} = 2\sqrt{14}.$$

Problem 2.2. Find the unit normal vector to the surface $\phi = x^2 + y^2 + z^2 - 3 = 0$ at $(1, 1, 1)$.

Solution. We know $\nabla\phi$ is the normal vector to the surface, so the unit normal vector is $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$ at $(1, 1, 1)$.

$$\nabla\phi = \vec{i} \frac{\partial}{\partial x}(x^2 + y^2 + z^2 - 3) + \vec{j} \frac{\partial}{\partial y}(\dots) + \vec{k} \frac{\partial}{\partial z}(\dots) = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}.$$

$$\text{At } (1, 1, 1): \nabla\phi = 2\vec{i} + 2\vec{j} + 2\vec{k}, \quad |\nabla\phi| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} = 2\sqrt{3}.$$

$$\hat{n} = \frac{1}{2\sqrt{3}} 2(\vec{i} + \vec{j} + \vec{k}) = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}.$$

Problem 2.3. Find the angle between the surfaces $x^2 + yz = 2$ and $x + 2y - z = 2$ at $(1, 1, 1)$.

Solution. Let $\phi_1 = x^2 + yz - 2$ and $\phi_2 = x + 2y - z - 2$.

$$\nabla\phi_1 = 2x\vec{i} + z\vec{j} + y\vec{k}, \quad \nabla\phi_2 = \vec{i} + 2\vec{j} - \vec{k}.$$

$$\text{At } (1, 1, 1): \nabla\phi_1 = 2\vec{i} + \vec{j} + \vec{k}, \quad |\nabla\phi_1| = \sqrt{4 + 1 + 1} = \sqrt{6}; \quad \nabla\phi_2 = \vec{i} + 2\vec{j} - \vec{k}, \quad |\nabla\phi_2| = \sqrt{1 + 4 + 1} = \sqrt{6}.$$

$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1| |\nabla\phi_2|} = \frac{(2\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + 2\vec{j} - \vec{k})}{\sqrt{6} \cdot \sqrt{6}} = \frac{2 + 2 - 1}{6} = \frac{3}{6} = \frac{1}{2},$$

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) \Rightarrow \theta = 60^\circ.$$

Problem 2.4. Find the angle between the surfaces $z = x^2 + y^2 - 3$ and $x^2 + y^2 + z^2 = 9$ at the point $(2, -1, 2)$.

Solution. Let $\phi_1 = x^2 + y^2 - 3 - z$ and $\phi_2 = x^2 + y^2 + z^2 - 9$.

$$\nabla\phi_1 = 2x\vec{i} + 2y\vec{j} - \vec{k}, \quad \nabla\phi_2 = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}.$$

At $(2, -1, 2)$:

$$\nabla\phi_1 = 4\vec{i} - 2\vec{j} - \vec{k}, \quad |\nabla\phi_1| = \sqrt{16 + 4 + 1} = \sqrt{21};$$

$$\nabla\phi_2 = 4\vec{i} - 2\vec{j} + 4\vec{k}, \quad |\nabla\phi_2| = \sqrt{16 + 4 + 16} = \sqrt{36} = 6.$$

$$\cos\theta = \frac{(4\vec{i} - 2\vec{j} - \vec{k}) \cdot (4\vec{i} - 2\vec{j} + 4\vec{k})}{\sqrt{21} \cdot 6} = \frac{16 + 4 - 4}{6\sqrt{21}} = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}, \quad \theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right).$$

Problem 2.5. Find the angle between the normals to the surface $xy^3z^2 = 4$ at the points $(-1, -1, 2)$ and $(4, 1, -1)$.

Solution. $\phi = xy^3z^2 - 4$. $\nabla\phi$ gives the normal to the surface:

$$\nabla\phi = y^3z^2\vec{i} + 3xy^2z^2\vec{j} + 2xy^3z\vec{k}.$$

At $(-1, -1, 2)$:

$$\nabla\phi = \vec{i}(-1)(4) + \vec{j}3(-1)(1)(4) + \vec{k}2(-1)(-1)(2) = -4\vec{i} - 12\vec{j} + 4\vec{k},$$

$$|\nabla\phi| = \sqrt{16 + 144 + 16} = \sqrt{176}.$$

At $(4, 1, -1)$:

$$\nabla\phi = \vec{i} + 12\vec{j} - 8\vec{k}, \quad |\nabla\phi| = \sqrt{1 + 144 + 64} = \sqrt{209}.$$

The angle between the normals is given by

$$\cos\theta = \frac{(-4\vec{i} - 12\vec{j} + 4\vec{k}) \cdot (\vec{i} + 12\vec{j} - 8\vec{k})}{\sqrt{176}\sqrt{209}} = \frac{-4 - 144 - 32}{\sqrt{176}\sqrt{209}} = \frac{-180}{\sqrt{176}\sqrt{209}}.$$

Taking the acute angle between the normal lines,

$$\theta = \cos^{-1}\left(\frac{180}{\sqrt{176}\sqrt{209}}\right) \approx 20.2^\circ.$$

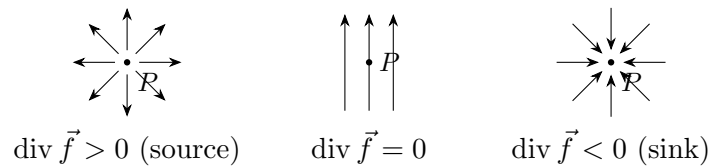
3 Divergence and Curl

Definition 3.1 (Divergence). If $\vec{f} = f_1\vec{i} + f_2\vec{j} + f_3\vec{k}$ is a vector-valued function, then the divergence of $\vec{f}$ is defined by

$$\operatorname{div} \vec{f} = \nabla \cdot \vec{f} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}\right) \cdot (f_1\vec{i} + f_2\vec{j} + f_3\vec{k}) = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}.$$

The result is a scalar-valued function.

The divergence of $\vec{f}$ is a scalar quantity; it tells us the rate at which the vector field spreads out from a particular point.



Definition 3.2 (Curl). If $\vec{f}$ is a vector-valued function, then its curl is defined by

$$\operatorname{curl} \vec{f} = \nabla \times \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \vec{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) - \vec{j} \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} \right) + \vec{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right).$$

The result is a vector-valued function.

The curl of $\vec{f}$ is a vector-valued function; it tells us the rate at which the vector field rotates about a particular point. A zero curl means no rotation.

- A vector field $\vec{f}$ is called **solenoidal** if $\operatorname{div} \vec{f} = 0$.
- A vector field $\vec{f}$ is called **irrotational** if $\operatorname{curl} \vec{f} = \vec{0}$.

3.1 Problems

Problem 3.1. Find the divergence and curl of the position vector $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

Solution.

$$\nabla \cdot \vec{r} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (x\vec{i} + y\vec{j} + z\vec{k}) = 1 + 1 + 1 \Rightarrow \boxed{\operatorname{div} \vec{r} = 3}.$$

$$\operatorname{curl} \vec{r} = \nabla \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \vec{i} \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) - \vec{j} \left(\frac{\partial z}{\partial x} - \frac{\partial x}{\partial z} \right) + \vec{k} \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right) \Rightarrow \boxed{\operatorname{curl} \vec{r} = \vec{0}}.$$

Problem 3.2. Prove that $\vec{f} = 3y^4z^2\vec{i} + 4x^3z^2\vec{j} - 3x^2y^2\vec{k}$ is solenoidal.

Solution.

$$\nabla \cdot \vec{f} = \frac{\partial}{\partial x}(3y^4z^2) + \frac{\partial}{\partial y}(4x^3z^2) + \frac{\partial}{\partial z}(-3x^2y^2) = 0 + 0 + 0 = 0.$$

Hence $\vec{f}$ is solenoidal.

Problem 3.3. Show that $\vec{f} = 2xy\vec{i} + (x^2 + 2yz)\vec{j} + (y^2 + 1)\vec{k}$ is irrotational.

Solution.

$$\nabla \times \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 + 2yz & y^2 + 1 \end{vmatrix} = \vec{i}(2y - 2y) - \vec{j}(0 - 0) + \vec{k}(2x - 2x) = 0\vec{i} + 0\vec{j} + 0\vec{k} = \vec{0}.$$

Hence $\vec{f}$ is irrotational.

Problem 3.4. Find a if $\vec{f} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + az)\vec{k}$ is solenoidal.

Solution. Since $\vec{f}$ is solenoidal, $\nabla \cdot \vec{f} = 0$:

$$\frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x + az) = 0 \Rightarrow 1 + 1 + a = 0 \Rightarrow \boxed{a = -2}.$$

Problem 3.5. Find a, b, c if $\vec{f} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational.

Solution.

$$\begin{aligned} \nabla \times \vec{f} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix} \\ &= \vec{i} \left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z) \right] - \vec{j} \left[\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az) \right] \\ &\quad + \vec{k} \left[\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x + 2y + az) \right] \\ &= (c + 1)\vec{i} - (4 - a)\vec{j} + (b - 2)\vec{k}. \end{aligned}$$

Since $\vec{f}$ is irrotational, $\nabla \times \vec{f} = \vec{0}$:

$$c + 1 = 0, \quad -4 + a = 0, \quad b - 2 = 0 \Rightarrow \boxed{a = 4, b = 2, c = -1}.$$

Problem 3.6. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ is the position vector, prove that $\text{div}(r^n \vec{r}) = (n + 3)r^n$. Also prove that $r^n \vec{r}$ is solenoidal if and only if $n = -3$.

Solution. $r^n \vec{r} = r^n x \vec{i} + r^n y \vec{j} + r^n z \vec{k}$. Then

$$\begin{aligned} \text{div}(r^n \vec{r}) &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (r^n x \vec{i} + r^n y \vec{j} + r^n z \vec{k}) = \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z) \\ &= nr^{n-1} \frac{\partial r}{\partial x} x + r^n + nr^{n-1} \frac{\partial r}{\partial y} y + r^n + nr^{n-1} \frac{\partial r}{\partial z} z + r^n. \end{aligned} \quad (1)$$

Now $r = \sqrt{x^2 + y^2 + z^2}$, i.e. $r^2 = x^2 + y^2 + z^2$. Differentiating with respect to x, y, z :

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

Substituting in (1):

$$\text{div}(r^n \vec{r}) = 3r^n + nr^{n-1} \left(\frac{x^2}{r} + \frac{y^2}{r} + \frac{z^2}{r} \right) = 3r^n + nr^{n-1} \frac{r^2}{r} = 3r^n + nr^n = (n + 3)r^n.$$

Hence $\text{div}(r^n \vec{r}) = (n + 3)r^n$.

$r^n \vec{r}$ is solenoidal $\iff \text{div}(r^n \vec{r}) = 0 \iff (n + 3)r^n = 0 \iff n = -3$.

Problem 3.7. Prove that $\text{curl}(r^n \vec{r}) = \vec{0}$, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

Solution.

$$\begin{aligned} \text{curl}(r^n \vec{r}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix} \\ &= \vec{i} \left(\frac{\partial}{\partial y}(r^n z) - \frac{\partial}{\partial z}(r^n y) \right) - \vec{j} \left(\frac{\partial}{\partial x}(r^n z) - \frac{\partial}{\partial z}(r^n x) \right) \\ &\quad + \vec{k} \left(\frac{\partial}{\partial x}(r^n y) - \frac{\partial}{\partial y}(r^n x) \right) \\ &= \vec{i} \left[nr^{n-1} \frac{\partial r}{\partial y} z - nr^{n-1} \frac{\partial r}{\partial z} y \right] - \vec{j} \left[nr^{n-1} \frac{\partial r}{\partial x} z - nr^{n-1} \frac{\partial r}{\partial z} x \right] \\ &\quad + \vec{k} \left[nr^{n-1} \frac{\partial r}{\partial x} y - nr^{n-1} \frac{\partial r}{\partial y} x \right]. \end{aligned}$$

Using $\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$:

$$\text{curl}(r^n \vec{r}) = \vec{i} nr^{n-1} \left[\frac{y}{r} z - \frac{z}{r} y \right] - \vec{j} nr^{n-1} \left[\frac{x}{r} z - \frac{z}{r} x \right] + \vec{k} nr^{n-1} \left[\frac{x}{r} y - \frac{y}{r} x \right] = nr^{n-1} \vec{0} = \vec{0}.$$

Problem 3.8. Prove that $\nabla^2(r^n \vec{r}) = n(n+3)r^{n-2}\vec{r}$, where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian operator.

Solution.

$$\nabla^2(r^n \vec{r}) = \frac{\partial^2}{\partial x^2}(r^n \vec{r}) + \frac{\partial^2}{\partial y^2}(r^n \vec{r}) + \frac{\partial^2}{\partial z^2}(r^n \vec{r}). \quad (1)$$

Using $\frac{\partial r}{\partial x} = \frac{x}{r}$ and $\frac{\partial \vec{r}}{\partial x} = \vec{i}$,

$$\frac{\partial}{\partial x}(r^n \vec{r}) = nr^{n-1} \frac{\partial r}{\partial x} \vec{r} + r^n \vec{i} = nr^{n-2} x \vec{r} + r^n \vec{i}.$$

Differentiating again with respect to x :

$$\begin{aligned} \frac{\partial^2}{\partial x^2}(r^n \vec{r}) &= n(n-2)r^{n-3} \frac{\partial r}{\partial x} x \vec{r} + nr^{n-2} \vec{r} + nr^{n-2} x \vec{i} + nr^{n-1} \frac{\partial r}{\partial x} \vec{i} \\ &= n(n-2)r^{n-4} x^2 \vec{r} + nr^{n-2} \vec{r} + 2nr^{n-2} x \vec{i}. \end{aligned} \quad (2)$$

Similarly,

$$\frac{\partial^2}{\partial y^2}(r^n \vec{r}) = n(n-2)r^{n-4} y^2 \vec{r} + nr^{n-2} \vec{r} + 2nr^{n-2} y \vec{j}, \quad (3)$$

$$\frac{\partial^2}{\partial z^2}(r^n \vec{r}) = n(n-2)r^{n-4} z^2 \vec{r} + nr^{n-2} \vec{r} + 2nr^{n-2} z \vec{k}. \quad (4)$$

Adding (2), (3), (4) and using $x^2 + y^2 + z^2 = r^2$, $x\vec{i} + y\vec{j} + z\vec{k} = \vec{r}$:

$$\begin{aligned} \nabla^2(r^n \vec{r}) &= n(n-2)r^{n-4}(x^2 + y^2 + z^2)\vec{r} + 3nr^{n-2}\vec{r} + 2nr^{n-2}(x\vec{i} + y\vec{j} + z\vec{k}) \\ &= n(n-2)r^{n-2}\vec{r} + 3nr^{n-2}\vec{r} + 2nr^{n-2}\vec{r} = [n^2 - 2n + 3n + 2n] r^{n-2}\vec{r} = n(n+3)r^{n-2}\vec{r}. \end{aligned}$$

4 Conservative Fields and Scalar Potential

Definition 4.1. A vector-valued function $\vec{f}$ is said to be a *conservative field* if $\text{curl } \vec{f} = \vec{0}$. If $\vec{f}$ is conservative, then we can find a scalar function ϕ such that

$$\vec{f} = \nabla \phi,$$

and ϕ is called the *scalar potential* (potential energy, or work done by the force $\vec{f}$).

4.1 Problems

Problem 4.1. Show that $\vec{f} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3z^2x\vec{k}$ is a conservative field and find its scalar potential.

Solution.

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3z^2x \end{vmatrix} = \vec{i}(0 - 0) - \vec{j}(3z^2 - 3z^2) + \vec{k}(2x - 2x) = \vec{0}.$$

Hence $\vec{f}$ is a conservative field. To find the scalar potential, let $\vec{f} = \nabla\phi$:

$$(2xy + z^3)\vec{i} + x^2\vec{j} + 3z^2x\vec{k} = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}.$$

Equating the $\vec{i}, \vec{j}, \vec{k}$ components and integrating partially:

$$\begin{aligned} \frac{\partial\phi}{\partial x} &= 2xy + z^3 & \frac{\partial\phi}{\partial y} &= x^2 & \frac{\partial\phi}{\partial z} &= 3z^2x \\ \phi &= x^2y + xz^3 + f(y, z) & \phi &= x^2y + g(x, z) & \phi &= xz^3 + h(x, y) \end{aligned}$$

Comparing all the above expressions for ϕ , we get

$$\boxed{\phi = x^2y + xz^3 + k},$$

where k is an arbitrary constant.

Problem 4.2. Show that $\vec{f} = 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$ is a conservative field and find its scalar potential ϕ such that $\phi(1, -2, 2) = 4$.

Solution.

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz^3 & x^2z^3 & 3x^2yz^2 \end{vmatrix} = \vec{i}(3x^2z^2 - 3x^2z^2) - \vec{j}(6xyz^2 - 6xyz^2) + \vec{k}(2xz^3 - 2xz^3) = \vec{0}.$$

So $\vec{f}$ is conservative. Let $\vec{f} = \nabla\phi$:

$$\begin{aligned} \frac{\partial\phi}{\partial x} &= 2xyz^3 & \frac{\partial\phi}{\partial y} &= x^2z^3 & \frac{\partial\phi}{\partial z} &= 3x^2yz^2 \\ \phi &= x^2yz^3 + f(y, z) & \phi &= x^2yz^3 + g(x, z) & \phi &= x^2yz^3 + h(x, y) \end{aligned}$$

Hence $\phi = x^2yz^3 + k$. Given $\phi(1, -2, 2) = 4$:

$$4 = (1)^2(-2)(2)^3 + k = -16 + k \Rightarrow k = 20.$$

$$\boxed{\phi = x^2yz^3 + 20}$$

5 Vector Identities

Problem 5.1. If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ is a constant vector and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, prove that
(i) $\text{div}(\vec{r} \times \vec{a}) = 0$; (ii) $\text{curl}(\vec{r} \times \vec{a}) = -2\vec{a}$.

Solution.

$$\vec{r} \times \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ a_1 & a_2 & a_3 \end{vmatrix} = \vec{i}(a_3y - a_2z) - \vec{j}(a_3x - a_1z) + \vec{k}(a_2x - a_1y).$$

(i)

$$\nabla \cdot (\vec{r} \times \vec{a}) = \frac{\partial}{\partial x}(a_3y - a_2z) - \frac{\partial}{\partial y}(a_3x - a_1z) + \frac{\partial}{\partial z}(a_2x - a_1y) = 0 + 0 + 0 = 0.$$

(ii)

$$\begin{aligned} \text{curl}(\vec{r} \times \vec{a}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_3y - a_2z & a_1z - a_3x & a_2x - a_1y \end{vmatrix} \\ &= \vec{i} \left(\frac{\partial}{\partial y}(a_2x - a_1y) - \frac{\partial}{\partial z}(a_1z - a_3x) \right) - \vec{j} \left(\frac{\partial}{\partial x}(a_2x - a_1y) - \frac{\partial}{\partial z}(a_3y - a_2z) \right) \\ &\quad + \vec{k} \left(\frac{\partial}{\partial x}(a_1z - a_3x) - \frac{\partial}{\partial y}(a_3y - a_2z) \right) \\ &= \vec{i}(-a_1 - a_1) - \vec{j}(a_2 + a_2) + \vec{k}(-a_3 - a_3) = -2a_1\vec{i} - 2a_2\vec{j} - 2a_3\vec{k} \\ &= -2(a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) = -2\vec{a}. \end{aligned}$$

Problem 5.2. Prove that $\text{div}(\text{curl } \vec{f}) = 0$.

Solution. Let $\vec{f} = f_1\vec{i} + f_2\vec{j} + f_3\vec{k}$. Then

$$\text{curl } \vec{f} = \vec{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \vec{j} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \vec{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right).$$

$$\begin{aligned} \text{div}(\text{curl } \vec{f}) &= \nabla \cdot (\nabla \times \vec{f}) = \frac{\partial}{\partial x} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \\ &= \frac{\partial^2 f_3}{\partial x \partial y} - \frac{\partial^2 f_2}{\partial x \partial z} + \frac{\partial^2 f_1}{\partial y \partial z} - \frac{\partial^2 f_3}{\partial y \partial x} + \frac{\partial^2 f_2}{\partial z \partial x} - \frac{\partial^2 f_1}{\partial z \partial y} = 0, \end{aligned}$$

since mixed partial derivatives are equal.

Problem 5.3. Prove that $\text{curl}(\text{grad } \phi) = \vec{0}$.

Solution. $\nabla \phi = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}$.

$$\begin{aligned} \nabla \times \nabla \phi &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} = \vec{i} \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) - \vec{j} \left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x} \right) + \vec{k} \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right) \\ &= \vec{0}. \end{aligned}$$

Problem 5.4. If $\phi = x^3 + y^3 + z^3 - 3xyz$, find (i) $\text{div}(\text{grad } \phi)$ and (ii) $\text{curl}(\text{grad } \phi)$.

Solution.

$$\text{grad } \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} = (3x^2 - 3yz) \vec{i} + (3y^2 - 3xz) \vec{j} + (3z^2 - 3xy) \vec{k}.$$

(i)

$$\text{div}(\text{grad } \phi) = \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy) = 6x + 6y + 6z.$$

(ii)

$$\text{curl}(\text{grad } \phi) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix} = \vec{i}(-3x+3x) - \vec{j}(-3y+3y) + \vec{k}(-3z+3z) = \vec{0}.$$

Unit IV: Vector Integration

6 Line Integrals

For a vector field $\vec{f} = f_1\vec{i} + f_2\vec{j} + f_3\vec{k}$ and a curve C ,

$$\int_C \vec{f} \cdot d\vec{r} = \int_C (f_1\vec{i} + f_2\vec{j} + f_3\vec{k}) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k}) = \int_C (f_1 dx + f_2 dy + f_3 dz).$$

We can then proceed with the ordinary line-integral technique to evaluate the integral.

Remark. 1. Generally, $\int_C \vec{f} \cdot d\vec{r}$ gives the *work done* by the force $\vec{f}$ in displacing a particle along the curve C from the point A to B .

2. If $\vec{f}$ is a conservative (irrotational) field, then $\vec{f}$ can be written as $\vec{f} = \nabla\phi$. In this case the work done by the force $\vec{f}$ does not depend on the path, but only on the end points A and B :

$$\int_C \vec{f} \cdot d\vec{r} = \int_C \nabla\phi \cdot d\vec{r} = \int_A^B \left(\frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \right) = \int_A^B d\phi = \phi(B) - \phi(A).$$

6.1 Problems

Problem 6.1. Find the work done when the force $\vec{f} = (x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}$ displaces a particle from $(0, 0)$ to $(1, 1)$ along the curve (i) $y = x$; (ii) $y^2 = x$.

Solution.

$$\int_C \vec{f} \cdot d\vec{r} = \int_C [(x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}] \cdot (dx\vec{i} + dy\vec{j}) = \int_C (x^2 - y^2 + x) dx - (2xy + y) dy.$$

(i) C : $y = x$ from $(0, 0)$ to $(1, 1)$, so $dy = dx$ and x varies from 0 to 1:

$$\begin{aligned} \int_C \vec{f} \cdot d\vec{r} &= \int_0^1 [(x^2 - x^2 + x) - (2x^2 + x)] dx \\ &= \int_0^1 (-2x^2) dx = -2 \left[\frac{x^3}{3} \right]_0^1 = -\frac{2}{3}. \end{aligned}$$

(ii) C : $y^2 = x$ from $(0, 0)$ to $(1, 1)$, so $dx = 2y dy$ and y varies from 0 to 1:

$$\begin{aligned} \int_C \vec{f} \cdot d\vec{r} &= \int_0^1 [(y^4 - y^2 + y^2)2y - (2y^3 + y)] dy = \int_0^1 (2y^5 - 2y^3 - y) dy \\ &= \left[\frac{2y^6}{6} - \frac{2y^4}{4} - \frac{y^2}{2} \right]_0^1 = \frac{1}{3} - \frac{1}{2} - \frac{1}{2} = -\frac{2}{3}. \end{aligned}$$

From the above we see that the value of $\int_C \vec{f} \cdot d\vec{r}$ does not depend on the path C ; $\vec{f}$ is a conservative field (irrotational vector).

Problem 6.2. Find the work done by the force $\vec{f} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k}$ when it moves a particle from $(1, -2, 1)$ to $(3, 1, 4)$ along any path.

Solution. The curve is not given, only the end points, so we check whether $\vec{f}$ is conservative:

$$\nabla \times \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = \vec{i}(0 - 0) - \vec{j}(3z^2 - 3z^2) + \vec{k}(2x - 2x) = \vec{0}.$$

$\therefore \vec{f}$ is a conservative field, and we can write $\vec{f} = \nabla\phi$:

$$\begin{aligned} \frac{\partial\phi}{\partial x} &= 2xy + z^3 & \frac{\partial\phi}{\partial y} &= x^2 & \frac{\partial\phi}{\partial z} &= 3xz^2 \\ \phi &= x^2y + xz^3 + f(y, z) & \phi &= x^2y + g(x, z) & \phi &= xz^3 + h(x, y) \end{aligned} \quad \Rightarrow \quad \phi = x^2y + xz^3.$$

$$\begin{aligned} \int_C \vec{f} \cdot d\vec{r} &= \int_C \nabla\phi \cdot d\vec{r} = \int_{(1,-2,1)}^{(3,1,4)} d\phi = \phi(3, 1, 4) - \phi(1, -2, 1) \\ &= [(3)^2(1) + 3(4)^3] - [(1)^2(-2) + 1(1)^3] = (9 + 192) - (-2 + 1) = 202. \end{aligned}$$

Problem 6.3. Find the work done by the force $\vec{f} = y(3x^2y - z^2)\vec{i} + x(2x^2y - z^2)\vec{j} - 2xyz\vec{k}$ when it moves a particle around a closed curve C .

Solution. Since no specific curve is given, let us check whether $\vec{f}$ is conservative. Here $\vec{f} = (3x^2y^2 - yz^2)\vec{i} + (2x^3y - xz^2)\vec{j} - 2xyz\vec{k}$.

$$\begin{aligned} \nabla \times \vec{f} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2y^2 - yz^2 & 2x^3y - xz^2 & -2xyz \end{vmatrix} \\ &= \vec{i}(-2xz + 2xz) - \vec{j}(-2yz + 2yz) + \vec{k}[(6x^2y - z^2) - (6x^2y - z^2)] = \vec{0}. \end{aligned}$$

So we can write $\vec{f} = \nabla\phi$:

$$\begin{aligned} \frac{\partial\phi}{\partial x} &= 3x^2y^2 - z^2y & \frac{\partial\phi}{\partial y} &= 2x^3y - z^2x & \frac{\partial\phi}{\partial z} &= -2xyz \\ \phi &= x^3y^2 - z^2yx + f(y, z) & \phi &= x^3y^2 - z^2yx + g(x, z) & \phi &= -xyz^2 + h(x, y) \end{aligned}$$

so $\phi = x^3y^2 - xyz^2$. For a closed curve the starting and ending points coincide, say at A :

$$\oint_C \vec{f} \cdot d\vec{r} = \oint_C \nabla\phi \cdot d\vec{r} = \int_A^A d\phi = \phi(A) - \phi(A) = 0.$$

7 Green's Theorem in the Plane

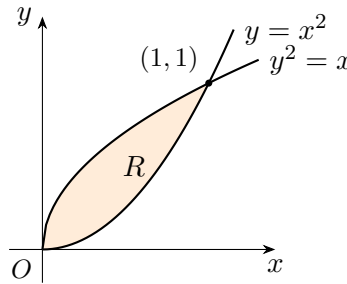
Green's theorem. Let $P(x, y)$ and $Q(x, y)$ be continuous functions defined inside and on a closed curve C bounding a region R , and let them have continuous first-order partial derivatives there. Then

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

7.1 Problems

Problem 7.1. Verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$, where C is the closed curve bounded by $y = x^2$ and $y^2 = x$.

Solution.



Here $P = 3x^2 - 8y^2$, $Q = 4y - 6xy$, so $\frac{\partial P}{\partial y} = -16y$, $\frac{\partial Q}{\partial x} = -6y$ and

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -6y + 16y = 10y. \quad (1)$$

L.H.S. $= \oint_C P dx + Q dy$, where $C = C_1 + C_2$.

Along C_1 ($y = x^2$, $dy = 2x dx$, x from 0 to 1):

$$\begin{aligned} \int_{C_1} &= \int_0^1 [(3x^2 - 8x^4) + (4x^2 - 6x^3)2x] dx = \int_0^1 (3x^2 + 8x^3 - 20x^4) dx \\ &= [x^3 + 2x^4 - 4x^5]_0^1 = 1 + 2 - 4 = -1. \end{aligned}$$

Along C_2 ($x = y^2$, $dx = 2y dy$, y from 1 to 0):

$$\begin{aligned} \int_{C_2} &= \int_1^0 [(3y^4 - 8y^2)2y + (4y - 6y^3)] dy = \int_1^0 (6y^5 - 22y^3 + 4y) dy \\ &= \left[y^6 - \frac{22y^4}{4} + 2y^2 \right]_1^0 = -\left(1 - \frac{11}{2} + 2 \right) = \frac{5}{2}. \end{aligned}$$

$$\text{L.H.S.} = -1 + \frac{5}{2} = \frac{3}{2}.$$

R.H.S. $= \iint_R 10y dy dx$ by (1). Drawing a vertical strip in the region, y varies from x^2 to $\sqrt{x}$ and x varies from 0 to 1:

$$\int_0^1 \int_{x^2}^{\sqrt{x}} 10y dy dx = \int_0^1 [5y^2]_{x^2}^{\sqrt{x}} dx = 5 \int_0^1 (x - x^4) dx$$

$$= 5 \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 = 5 \left(\frac{1}{2} - \frac{1}{5} \right) = 5 \cdot \frac{3}{10} = \frac{3}{2}.$$

L.H.S. = R.H.S.; hence Green's theorem is verified.

Problem 7.2. Using Green's theorem, evaluate $\oint_C x^2(1+y) dx + (y^3 + x^3) dy$, where C is the boundary of the square $x = \pm 1, y = \pm 1$.

Solution. $P = x^2(1+y) = x^2 + x^2y, Q = y^3 + x^3; \quad \frac{\partial P}{\partial y} = x^2, \frac{\partial Q}{\partial x} = 3x^2.$

$$\begin{aligned} I &= \iint_R (3x^2 - x^2) dx dy = \int_{-1}^1 \int_{-1}^1 2x^2 dx dy = \int_{-1}^1 \left[\frac{2x^3}{3} \right]_{-1}^1 dy \\ &= \frac{2}{3} \int_{-1}^1 (1+1) dy = \frac{4}{3} [y]_{-1}^1 = \frac{8}{3}. \end{aligned}$$

Problem 7.3. Show by Green's theorem that $\frac{1}{2} \oint_C (x dy - y dx)$ is the area bounded by any simple closed curve C , and hence find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution. $I = \frac{1}{2} \oint_C x dy - y dx = \oint_C P dx + Q dy$ with $P = -\frac{y}{2}, Q = \frac{x}{2}$. Then $\frac{\partial P}{\partial y} = -\frac{1}{2}, \frac{\partial Q}{\partial x} = \frac{1}{2}$, so $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$. By Green's theorem,

$$I = \iint_R 1 dx dy = \text{area of the region } R \text{ bounded by } C.$$

Area of the ellipse. Using the parametric form $x = a \cos \theta, y = b \sin \theta$ ($dx = -a \sin \theta d\theta, dy = b \cos \theta d\theta$), θ from 0 to 2π :

$$\text{Area} = \frac{1}{2} \int_0^{2\pi} [a \cos \theta (b \cos \theta) - b \sin \theta (-a \sin \theta)] d\theta = \frac{ab}{2} \int_0^{2\pi} (\cos^2 \theta + \sin^2 \theta) d\theta = \frac{ab}{2} [\theta]_0^{2\pi} = \pi ab.$$

Problem 7.4. Using the above result, find the area of the circle $x^2 + y^2 = a^2$.

Solution. Parametric form: $x = a \cos \theta, y = a \sin \theta; dx = -a \sin \theta d\theta, dy = a \cos \theta d\theta$.

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \int_0^{2\pi} (a^2 \cos^2 \theta + a^2 \sin^2 \theta) d\theta = \frac{a^2}{2} [\theta]_0^{2\pi} = \pi a^2.$$

Problem 7.5. Verify Green's theorem for $\oint_C (x - 2y) dx + x dy$ taken around the circle $x^2 + y^2 = 1$.

Solution. $P = x - 2y$, $Q = x$; $\frac{\partial P}{\partial y} = -2$, $\frac{\partial Q}{\partial x} = 1$, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3$.

L.H.S. On C : $x = \cos \theta$, $y = \sin \theta$, $dx = -\sin \theta d\theta$, $dy = \cos \theta d\theta$, θ from 0 to 2π . Using $\sin 2\theta = 2 \sin \theta \cos \theta$, $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$, $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$:

$$\begin{aligned}\oint_C &= \int_0^{2\pi} [(\cos \theta - 2 \sin \theta)(-\sin \theta) + \cos \theta \cos \theta] d\theta = \int_0^{2\pi} (-\cos \theta \sin \theta + 2 \sin^2 \theta + \cos^2 \theta) d\theta \\ &= \int_0^{2\pi} \left[-\frac{\sin 2\theta}{2} + (1 - \cos 2\theta) + \frac{1 + \cos 2\theta}{2} \right] d\theta = \frac{1}{2} \int_0^{2\pi} (-\sin 2\theta + 3 - \cos 2\theta) d\theta \\ &= \frac{1}{2} \left[\frac{\cos 2\theta}{2} + 3\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} = \frac{1}{2} [0 + 6\pi - 0] = 3\pi.\end{aligned}$$

R.H.S. $= \iint_R 3 dx dy = 3 (\text{area of the unit circle}) = 3\pi$.

(Or) By symmetry, R.H.S. $= 4 \iint_{R_1} 3 dx dy$ over the first quadrant, where y varies from 0 to $\sqrt{1 - x^2}$ and x from 0 to 1:

$$12 \int_0^1 \int_0^{\sqrt{1-x^2}} dy dx = 12 \int_0^1 \sqrt{1-x^2} dx.$$

Put $x = \cos \theta$, $dx = -\sin \theta d\theta$ ($x = 0 \Rightarrow \theta = \frac{\pi}{2}$, $x = 1 \Rightarrow \theta = 0$):

$$12 \int_{\pi/2}^0 \sin \theta (-\sin \theta) d\theta = 12 \int_0^{\pi/2} \sin^2 \theta d\theta = 6 \int_0^{\pi/2} (1 - \cos 2\theta) d\theta = 6 \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/2} = 6 \cdot \frac{\pi}{2} = 3\pi.$$

L.H.S. = R.H.S. = 3π ; Green's theorem is verified.

Homework

1. Verify Green's theorem for $\oint_C (x^2 + y^2) dx + (y + 2x) dy$, where C is the boundary of the region enclosed by the curves $y = x^2$ and $y^2 = x$.
2. Using Green's theorem, evaluate $\oint_C (xy - x^2) dx + x^2 y dy$, where C is the closed curve enclosed by the lines $y = 0$, $x = 1$ and $y = x$.
3. Find the area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ using Green's theorem, $\frac{1}{2} \oint_C x dy - y dx$.

Problem 7.6. Verify Green's theorem for $\oint_C (x^2 + y^2) dx + (y + 2x) dy$, where C is the boundary of the region enclosed by $y = x^2$ and $y^2 = x$.

Solution. $P = x^2 + y^2$, $Q = y + 2x$; $\frac{\partial P}{\partial y} = 2y$, $\frac{\partial Q}{\partial x} = 2$, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2 - 2y$.

L.H.S. Along C_1 ($y = x^2$, $dy = 2x dx$, x from 0 to 1):

$$\int_{C_1} = \int_0^1 [(x^2 + x^4) + (x^2 + 2x)2x] dx$$

$$= \int_0^1 (5x^2 + 2x^3 + x^4) dx = \frac{5}{3} + \frac{1}{2} + \frac{1}{5} = \frac{71}{30}.$$

Along C_2 ($x = y^2$, $dx = 2y dy$, y from 1 to 0):

$$\begin{aligned} \int_{C_2} &= \int_1^0 [(y^4 + y^2)2y + (y + 2y^2)] dy = \int_1^0 (2y^5 + 2y^3 + y + 2y^2) dy \\ &= \left[\frac{y^6}{3} + \frac{y^4}{2} + \frac{y^2}{2} + \frac{2y^3}{3} \right]_1^0 = -\frac{1}{3} - \frac{1}{2} - \frac{1}{2} - \frac{2}{3} = -2. \end{aligned}$$

$$\text{L.H.S.} = \frac{71}{30} - 2 = \frac{11}{30}.$$

R.H.S. (y from x^2 to $\sqrt{x}$, x from 0 to 1):

$$\begin{aligned} \int_0^1 \int_{x^2}^{\sqrt{x}} (2 - 2y) dy dx &= 2 \int_0^1 \left[y - \frac{y^2}{2} \right]_{x^2}^{\sqrt{x}} dx = 2 \int_0^1 \left(\sqrt{x} - \frac{x}{2} - x^2 + \frac{x^4}{2} \right) dx \\ &= 2 \left[\frac{2}{3} x^{3/2} - \frac{x^2}{4} - \frac{x^3}{3} + \frac{x^5}{10} \right]_0^1 = 2 \left[\frac{2}{3} - \frac{1}{4} - \frac{1}{3} + \frac{1}{10} \right] = 2 \cdot \frac{11}{60} = \frac{11}{30}. \end{aligned}$$

L.H.S. = R.H.S.; hence proved.

Problem 7.7. Using Green's theorem, evaluate $\oint_C (xy - x^2) dx + x^2y dy$, where C is enclosed by $y = 0$, $x = 1$ and $y = x$.

Solution. $P = xy - x^2$, $Q = x^2y$; $\frac{\partial P}{\partial y} = x$, $\frac{\partial Q}{\partial x} = 2xy$. The region is the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$: y from 0 to x , x from 0 to 1.

$$I = \int_0^1 \int_0^x (2xy - x) dy dx = \int_0^1 x [y^2 - y]_0^x dx = \int_0^1 x(x^2 - x) dx = \int_0^1 (x^3 - x^2) dx = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

Problem 7.8. Find the area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ using $\frac{1}{2} \oint_C x dy - y dx$.

Solution. Parametric form: $x = a \cos^3 t$, $y = a \sin^3 t$, $0 \leq t \leq 2\pi$; $dx = -3a \cos^2 t \sin t dt$, $dy = 3a \sin^2 t \cos t dt$.

$$x dy - y dx = 3a^2 (\cos^4 t \sin^2 t + \sin^4 t \cos^2 t) dt = 3a^2 \sin^2 t \cos^2 t dt = \frac{3a^2}{4} \sin^2 2t dt.$$

$$\text{Area} = \frac{1}{2} \int_0^{2\pi} \frac{3a^2}{4} \sin^2 2t dt = \frac{3a^2}{8} \int_0^{2\pi} \frac{1 - \cos 4t}{2} dt = \frac{3a^2}{8} \cdot \pi = \frac{3\pi a^2}{8}.$$

8 Surface Integrals and Stokes' Theorem

If $\vec{f}$ is a vector-valued function defined on a surface S whose projection is the region R , then

$$\iint_S \vec{f} \cdot d\vec{s} = \iint_S \vec{f} \cdot \hat{n} ds.$$

When S is a plane face (as for a cube or cuboid), ds is simply $dx dy$, $dy dz$ or $dz dx$. When S is a curved surface (like a sphere), we project it onto a coordinate plane:

$$\iint_S \vec{f} \cdot \hat{n} ds = \iint_R \frac{\vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{k}|} dx dy = \iint_R \frac{\vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{j}|} dy dz = \iint_R \frac{\vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{i}|} dz dx,$$

where R is the projection of S on the xy -, yz - or zx -plane respectively.

Note. The total flux of $\vec{f}$ across the surface S is given by the above integral.

Stokes' theorem. If S is an open two-sided surface bounded by a closed curve C , and $\vec{f}$ is a vector-valued function with continuous first-order partial derivatives on S , then

$$\oint_C \vec{f} \cdot d\vec{r} = \iint_S (\text{curl } \vec{f}) \cdot \hat{n} ds.$$

8.1 Problems

Problem 8.1. Evaluate by Stokes' theorem $\oint_C (yz dx + zx dy + xy dz)$, where C is the curve $x^2 + y^2 = 1$, $z = y^2$.

Solution. $I = \oint_C (yz \vec{i} + zx \vec{j} + xy \vec{k}) \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k}) = \oint_C \vec{f} \cdot d\vec{r}$ with $\vec{f} = yz \vec{i} + zx \vec{j} + xy \vec{k}$.
By Stokes' theorem

$$I = \iint_S \text{curl } \vec{f} \cdot \hat{n} ds. \quad (1)$$

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & zx & xy \end{vmatrix} = (x - x)\vec{i} - (y - y)\vec{j} + (z - z)\vec{k} = \vec{0}.$$

From (1), $I = \iint_S \vec{0} \cdot \hat{n} ds = 0$.

Problem 8.2. Verify Stokes' theorem for $\vec{f} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$, where S is the upper half of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

Solution. We verify $\oint_C \vec{f} \cdot d\vec{r} = \iint_S \text{curl } \vec{f} \cdot \hat{n} ds$. Here C is the circle $x^2 + y^2 = 1$ in the plane $z = 0$.

L.H.S.

$$\oint_C \vec{f} \cdot d\vec{r} = \oint_C (2x - y) dx - yz^2 dy - y^2z dz = \oint_C (2x - y) dx \quad (z = 0).$$

With $x = \cos \theta$, $y = \sin \theta$, $dx = -\sin \theta d\theta$, θ from 0 to 2π :

$$\oint_C = \int_0^{2\pi} (2 \cos \theta - \sin \theta)(-\sin \theta) d\theta = \int_0^{2\pi} (-2 \sin \theta \cos \theta + \sin^2 \theta) d\theta$$

$$= \int_0^{2\pi} \left(-\sin 2\theta + \frac{1 - \cos 2\theta}{2} \right) d\theta = \left[\frac{\cos 2\theta}{2} \right]_0^{2\pi} + \frac{1}{2} [\theta]_0^{2\pi} - \frac{1}{4} [\sin 2\theta]_0^{2\pi} = \pi.$$

R.H.S. S is a curved surface, so

$$\iint_S \text{curl } \vec{f} \cdot \hat{n} \, ds = \iint_R \frac{\text{curl } \vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{k}|} \, dx \, dy, \quad (2)$$

where R is the projection of S on the xy -plane: $x^2 + y^2 \leq 1$.

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix} = \vec{i}(-2yz + 2yz) - \vec{j}(0) + \vec{k}(0 + 1) = \vec{k}.$$

For $\phi = x^2 + y^2 + z^2 - 1$: $\nabla\phi = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}$, and

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\vec{i} + 2y\vec{j} + 2z\vec{k}}{2\sqrt{x^2 + y^2 + z^2}} = x\vec{i} + y\vec{j} + z\vec{k} \quad (x^2 + y^2 + z^2 = 1).$$

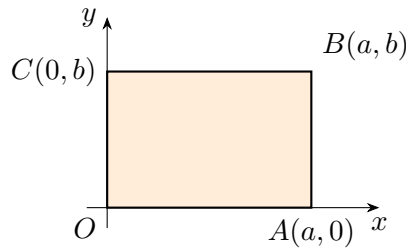
$\text{curl } \vec{f} \cdot \hat{n} = z$ and $\hat{n} \cdot \vec{k} = z$, so $\frac{\text{curl } \vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{k}|} = 1$. From (2),

$$\text{R.H.S.} = \iint_R 1 \, dx \, dy = 4 \int_0^1 \int_0^{\sqrt{1-y^2}} dx \, dy = \pi \quad (\text{area of the unit circle}).$$

$\therefore \text{L.H.S.} = \text{R.H.S.} = \pi$.

Problem 8.3. Verify Stokes' theorem for $\vec{f} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ in the rectangular region bounded by $x = 0$, $y = 0$, $x = a$ and $y = b$.

Solution.



$$\vec{f} \cdot d\vec{r} = (x^2 - y^2) dx + 2xy dy.$$

$$\text{L.H.S.} = (\int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}) \vec{f} \cdot d\vec{r}.$$

- OA : $y = 0$, $dy = 0$, x from 0 to a : $\int_0^a x^2 dx = \frac{a^3}{3}$.
- AB : $x = a$, $dx = 0$, y from 0 to b : $\int_0^b 2ay dy = [ay^2]_0^b = ab^2$.
- BC : $y = b$, $dy = 0$, x from a to 0: $\int_a^0 (x^2 - b^2) dx = \left[\frac{x^3}{3} - b^2x \right]_a^0 = ab^2 - \frac{a^3}{3}$.
- CO : $x = 0$, $dx = 0$, y from b to 0: $\int_{CO} \vec{f} \cdot d\vec{r} = 0$.

$$\oint_C \vec{f} \cdot d\vec{r} = \frac{a^3}{3} + ab^2 + ab^2 - \frac{a^3}{3} + 0 = 2ab^2.$$

$$\text{R.H.S.}^1 \text{ curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 & 2xy & 0 \end{vmatrix} = (2y + 2y)\vec{k} = 4y\vec{k}, \text{ and } \hat{n} = \vec{k} \text{ on the rectangle:}$$

$$\iint_S \text{curl } \vec{f} \cdot \hat{n} \, ds = \int_0^a \int_0^b 4y \, dy \, dx = \int_0^a 2b^2 \, dx = 2ab^2.$$

L.H.S. = R.H.S.; Stokes' theorem is verified.

Problem 8.4. Apply Stokes' theorem to evaluate $\oint_C 4y \, dx + 2z \, dy + 6y \, dz$, where C is the curve of intersection of the sphere $x^2 + y^2 + z^2 = 6z$ and the plane $z = x + 3$.

Solution. $I = \oint_C (4y\vec{i} + 2z\vec{j} + 6y\vec{k}) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k}) = \oint_C \vec{f} \cdot d\vec{r}$ with $\vec{f} = 4y\vec{i} + 2z\vec{j} + 6y\vec{k}$. By Stokes' theorem,

$$I = \iint_S \text{curl } \vec{f} \cdot \hat{n} \, ds, \quad (1)$$

where S is the plane surface bounded by C .

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & 2z & 6y \end{vmatrix} = \vec{i}(6 - 2) - \vec{j}(0 - 0) + \vec{k}(0 - 4) = 4\vec{i} - 4\vec{k}.$$

For the plane $\phi = z - x - 3$: $\nabla\phi = -\vec{i} + \vec{k}$, so $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{\sqrt{2}}(-\vec{i} + \vec{k})$. Hence

$$\text{curl } \vec{f} \cdot \hat{n} = (4\vec{i} - 4\vec{k}) \cdot \frac{1}{\sqrt{2}}(-\vec{i} + \vec{k}) = \frac{-4 - 4}{\sqrt{2}} = -4\sqrt{2},$$

a constant. So (1) becomes

$$I = -4\sqrt{2} \iint_S ds = -4\sqrt{2} \times (\text{area of the circle } C). \quad (2)$$

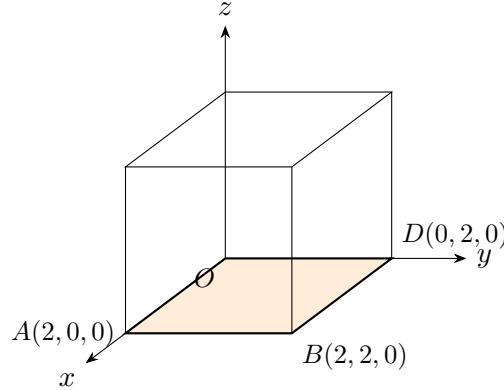
Now the sphere is $x^2 + y^2 + (z - 3)^2 = 9$, with centre $(0, 0, 3)$ and radius 3. The plane $z = x + 3$ passes through the centre $(0, 0, 3)$, so C is a great circle of radius 3. From (2),

$$I = -4\sqrt{2} \times \pi(3)^2 = -36\sqrt{2}\pi.$$

¹The handwritten notes end after the L.H.S.; the R.H.S. is completed here.

Problem 8.5. Verify Stokes' theorem for $\vec{f} = (y - z + 2)\vec{i} + (yz + 4)\vec{j} - xz\vec{k}$ over the surface of the cube bounded by $x = 0$ to 2, $y = 0$ to 2 and $z = 0$ to 2, above the xy -plane (open at the bottom).

Solution.



The open surface consists of five faces $S_1, \dots, S_5$; its boundary C is the square $OABD$ in the xy -plane. We verify

$$\oint_{OABD} \vec{f} \cdot d\vec{r} = \left(\iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} \right) \text{curl } \vec{f} \cdot \hat{n} \, ds.$$

L.H.S. On $z = 0$ ($dz = 0$): $\vec{f} \cdot d\vec{r} = (y - z + 2) dx + (yz + 4) dy - xz dz = (y + 2) dx + 4 dy$.

- OA : $O(0,0,0) \rightarrow A(2,0,0)$, $y = 0$, $dy = 0$: $\int_0^2 2 dx = [2x]_0^2 = 4$.
- AB : $A(2,0,0) \rightarrow B(2,2,0)$, $x = 2$, $dx = 0$: $\int_0^2 4 dy = [4y]_0^2 = 8$.
- BD : $B(2,2,0) \rightarrow D(0,2,0)$, $y = 2$, $dy = 0$: $\int_2^0 (2 + 2) dx = [4x]_2^0 = -8$.
- DO : $D(0,2,0) \rightarrow O$, $x = 0$, $dx = 0$: $\int_2^0 4 dy = [4y]_2^0 = -8$.

$$\text{L.H.S.} = 4 + 8 - 8 - 8 = -4.$$

R.H.S.

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z + 2 & yz + 4 & -xz \end{vmatrix} = (0 - y)\vec{i} - \vec{j}(-z + 1) - \vec{k} = -y\vec{i} + (z - 1)\vec{j} - \vec{k}.$$

- S_1 : the xz -plane, $y = 0$, $\hat{n} = -\vec{j}$; x, z from 0 to 2. $\text{curl } \vec{f} \cdot \hat{n} = 1 - z$:

$$\iint_{S_1} = \int_0^2 \int_0^2 (1 - z) dz dx = \int_0^2 \left[z - \frac{z^2}{2} \right]_0^2 dx = \int_0^2 0 dx = 0.$$

- S_2 : parallel to the xz -plane, $y = 2$, $\hat{n} = \vec{j}$. $\text{curl } \vec{f} \cdot \hat{n} = z - 1$:

$$\iint_{S_2} = \int_0^2 \int_0^2 (z - 1) dz dx = \int_0^2 \left[\frac{(z - 1)^2}{2} \right]_0^2 dx = \int_0^2 \frac{1}{2}(1 - 1) dx = 0.$$

- S_3 : the yz -plane, $x = 0$, $\hat{n} = -\vec{i}$. $\text{curl } \vec{f} \cdot \hat{n} = y$:

$$\iint_{S_3} = \int_0^2 \int_0^2 y dy dz = \int_0^2 \left[\frac{y^2}{2} \right]_0^2 dz = \int_0^2 2 dz = 4.$$

- S_4 : parallel to the yz -plane, $x = 2$, $\hat{n} = \vec{i}$. $\text{curl } \vec{f} \cdot \hat{n} = -y$: $\iint_{S_4} = -\int_0^2 \int_0^2 y \, dy \, dz = -4$.
- S_5 : the top face, parallel to the xy -plane, $z = 2$, $\hat{n} = \vec{k}$. $\text{curl } \vec{f} \cdot \hat{n} = -1$:

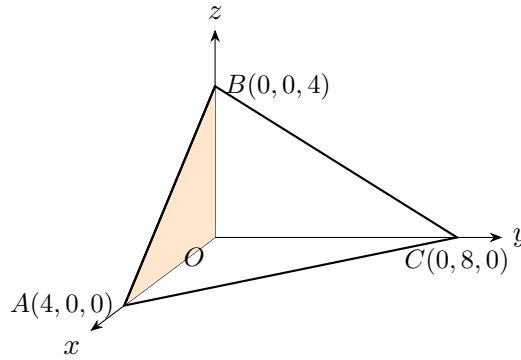
$$\iint_{S_5} = -\int_0^2 \int_0^2 dx \, dy = -\int_0^2 [x]_0^2 \, dy = -4.$$

$$\text{R.H.S.} = 0 + 0 + 4 - 4 - 4 = -4.$$

L.H.S. = R.H.S.; hence Stokes' theorem is verified.

Problem 8.6. Verify Stokes' theorem for $\vec{f} = xz\vec{i} + y\vec{j} + x^2y\vec{k}$, where S is the surface of the region bounded by $x = 0$, $y = 0$, $z = 0$ and $2x + y + 2z = 8$, which is not included in the xz -plane.

Solution.



The plane $2x + y + 2z = 8$, i.e. $\frac{x}{4} + \frac{y}{8} + \frac{z}{4} = 1$, meets the axes at $A(4, 0, 0)$, $C(0, 8, 0)$, $B(0, 0, 4)$. The face OAB in the xz -plane is excluded, so the boundary C of S is the triangle OAB in the xz -plane ($y = 0$, $dy = 0$). We verify

$$\left(\int_{AO} + \int_{OB} + \int_{BA} \right) \vec{f} \cdot d\vec{r} = \left(\iint_{S_1} + \iint_{S_2} + \iint_{S_3} \right) \text{curl } \vec{f} \cdot \hat{n} \, ds.$$

L.H.S. With $y = 0$: $\vec{f} \cdot d\vec{r} = xz \, dx + y \, dy + x^2y \, dz = xz \, dx$.

- AO : $y = 0$, $z = 0$, x from 4 to 0: $\int_4^0 x \cdot 0 \, dx = 0$.
- OB : $x = 0$, $y = 0$, z from 0 to 4: $\int_0^4 x^2y \, dz = 0$.
- BA : $B(0, 0, 4) \rightarrow A(4, 0, 0)$, $y = 0$, $z = 4 - x$, x from 0 to 4:

$$\int_0^4 x(4 - x) \, dx = \left[2x^2 - \frac{x^3}{3} \right]_0^4 = 32 - \frac{64}{3} = \frac{32}{3}.$$

$$\text{L.H.S.} = 0 + 0 + \frac{32}{3} = \frac{32}{3}.$$

R.H.S.

$$\text{curl } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & y & x^2y \end{vmatrix} = \vec{i}(x^2 - 0) - \vec{j}(2xy - x) + \vec{k}(0 - 0) = x^2\vec{i} + (x - 2xy)\vec{j}.$$

- S_1 : in the xy -plane, $\hat{n} = -\vec{k}$: $\text{curl } \vec{f} \cdot \hat{n} = 0$, so $\iint_{S_1} = 0$.

- S_2 : in the yz -plane ($x = 0$), $\hat{n} = -\vec{i}$: $\text{curl } \vec{f} \cdot \hat{n} = -x^2 = 0$, so $\iint_{S_2} = 0$.
- S_3 : the slant face $2x + y + 2z = 8$. Project onto the xy -plane: $z = 0$ gives $2x + y = 8$, so y varies from 0 to $8 - 2x$ and x from 0 to 4. With $\phi = 2x + y + 2z - 8$,

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2\vec{i} + \vec{j} + 2\vec{k}}{3}, \quad \text{curl } \vec{f} \cdot \hat{n} = \frac{1}{3} [2x^2 + x - 2xy], \quad |\hat{n} \cdot \vec{k}| = \frac{2}{3},$$

$$\frac{\text{curl } \vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{k}|} = x^2 + \frac{x}{2} - xy.$$

$$\begin{aligned} \iint_{S_3} &= \int_0^4 \int_0^{8-2x} \left(x^2 + \frac{x}{2} - xy \right) dy dx = \int_0^4 \left[\left(x^2 + \frac{x}{2} \right) (8 - 2x) - \frac{x(8 - 2x)^2}{2} \right] dx \\ &= \int_0^4 (-4x^3 + 23x^2 - 28x) dx = \left[-x^4 + \frac{23x^3}{3} - 14x^2 \right]_0^4 = -256 + \frac{1472}{3} - 224 = \frac{32}{3}. \end{aligned}$$

$$\text{R.H.S.} = 0 + 0 + \frac{32}{3} = \frac{32}{3} = \text{L.H.S.}; \text{ Stokes' theorem is verified.}$$

9 Gauss Divergence Theorem

Gauss divergence theorem. If $\vec{f}$ is a vector-valued function defined on a closed surface S bounding the volume V , then

$$\iint_S \vec{f} \cdot \hat{n} ds = \iiint_V \text{div } \vec{f} dv.$$

The value of the above integral gives the total amount of spread (flux) of $\vec{f}$ over the surface S .

9.1 Problems

Problem 9.1. Use the Gauss divergence theorem to evaluate $\iint_S \vec{f} \cdot \hat{n} ds$, where $\vec{f} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and S is the surface of the cylinder $x^2 + y^2 = 4$, $z = 0$ to $z = 3$.

Solution. By the Gauss divergence theorem, $\iint_S \vec{f} \cdot \hat{n} ds = \iiint_V \text{div } \vec{f} dv$.

$$\text{div } \vec{f} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}) = 4 - 4y + 2z.$$

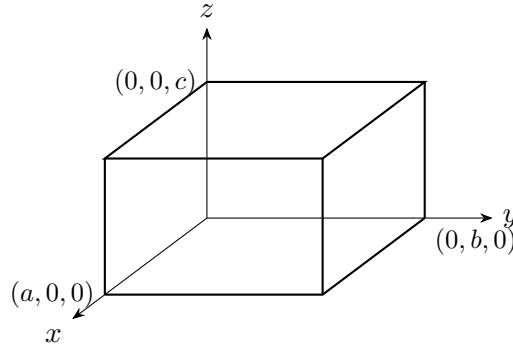
In the cylinder, y varies from $-\sqrt{4-x^2}$ to $\sqrt{4-x^2}$, x from -2 to 2 , and z from 0 to 3 .

$$\begin{aligned} \iiint_V \text{div } \vec{f} dv &= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) dz dy dx = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} [4z - 4yz + z^2]_0^3 dy dx \\ &= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y) dy dx = \int_{-2}^2 [21y - 6y^2]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \\ &= \int_{-2}^2 \left[21 \left(2\sqrt{4-x^2} \right) - 6(0) \right] dx = 42 \int_{-2}^2 \sqrt{2^2 - x^2} dx \end{aligned}$$

$$\begin{aligned}
&= 42 \cdot 2 \int_0^2 \sqrt{2^2 - x^2} dx \quad (\text{even function}) \\
&= 84 \left[\frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2 = 84 \left[0 + 2 \cdot \frac{\pi}{2} - 0 \right] = 84\pi.
\end{aligned}$$

Problem 9.2. Verify the Gauss divergence theorem for $\vec{f} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ taken over the rectangular parallelepiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$.

Solution.



$$\text{L.H.S.} = \iint_S \vec{f} \cdot \hat{n} ds = \left(\iint_{S_1} + \dots + \iint_{S_6} \right) \vec{f} \cdot \hat{n} ds.$$

- S_1 : the xy -plane, $z = 0$, $\hat{n} = -\vec{k}$: $\vec{f} \cdot \hat{n} = -(z^2 - xy) = xy$.

$$\iint_{S_1} = \int_0^a \int_0^b xy dy dx = \int_0^a x \frac{b^2}{2} dx = \frac{a^2 b^2}{4}.$$

- S_2 : parallel to the xy -plane, $z = c$, $\hat{n} = \vec{k}$: $\vec{f} \cdot \hat{n} = c^2 - xy$.

$$\iint_{S_2} = \int_0^a \int_0^b (c^2 - xy) dy dx = \int_0^a \left(c^2 b - x \frac{b^2}{2} \right) dx = abc^2 - \frac{a^2 b^2}{4}.$$

- S_3 : the xz -plane, $y = 0$, $\hat{n} = -\vec{j}$: $\vec{f} \cdot \hat{n} = -(y^2 - zx) = zx$.

$$\iint_{S_3} = \int_0^a \int_0^c zx dz dx = \int_0^a \frac{c^2}{2} x dx = \frac{a^2 c^2}{4}.$$

- S_4 : parallel to the xz -plane, $y = b$, $\hat{n} = \vec{j}$: $\vec{f} \cdot \hat{n} = b^2 - zx$.

$$\iint_{S_4} = \int_0^a \int_0^c (b^2 - zx) dz dx = \int_0^a \left(b^2 c - \frac{c^2}{2} x \right) dx = ab^2 c - \frac{a^2 c^2}{4}.$$

- S_5 : the yz -plane, $x = 0$, $\hat{n} = -\vec{i}$: $\vec{f} \cdot \hat{n} = -(x^2 - yz) = yz$.

$$\iint_{S_5} = \int_0^b \int_0^c yz dz dy = \int_0^b \frac{c^2}{2} y dy = \frac{b^2 c^2}{4}.$$

- S_6 : parallel to the yz -plane, $x = a$, $\hat{n} = \vec{i}$: $\vec{f} \cdot \hat{n} = a^2 - yz$.

$$\iint_{S_6} = \int_0^b \int_0^c (a^2 - yz) dz dy = \int_0^b \left(a^2 c - y \frac{c^2}{2} \right) dy = a^2 bc - \frac{b^2 c^2}{4}.$$

Adding,

$$\begin{aligned} \text{L.H.S.} &= \frac{a^2b^2}{4} + abc^2 - \frac{a^2b^2}{4} + \frac{a^2c^2}{4} + ab^2c - \frac{a^2c^2}{4} \\ &+ \frac{b^2c^2}{4} + a^2bc - \frac{b^2c^2}{4} = abc^2 + ab^2c + a^2bc = abc(a + b + c). \end{aligned}$$

$$\text{R.H.S. } \operatorname{div} \vec{f} = 2x + 2y + 2z.$$

$$\begin{aligned} \iiint_V \operatorname{div} \vec{f} dv &= 2 \int_0^a \int_0^b \int_0^c (x + y + z) dz dy dx = 2 \int_0^a \int_0^b \left(cx + cy + \frac{c^2}{2} \right) dy dx \\ &= 2 \int_0^a \left(bcx + \frac{b^2c}{2} + \frac{bc^2}{2} \right) dx = 2 \left[\frac{a^2bc}{2} + \frac{ab^2c}{2} + \frac{abc^2}{2} \right] \\ &= a^2bc + ab^2c + abc^2 = abc(a + b + c). \end{aligned}$$

L.H.S. = R.H.S.; hence the theorem is verified.

9.2 Scalar form of the divergence theorem

We know from the Gauss divergence theorem that

$$\iint_S \vec{f} \cdot \hat{n} ds = \iiint_V \operatorname{div} \vec{f} dv, \quad (1)$$

and also

$$\iint_S \vec{f} \cdot d\vec{s} = \iint_S \vec{f} \cdot \hat{n} ds = \iint_R \frac{\vec{f} \cdot \hat{n}}{|\hat{n} \cdot \vec{k}|} dx dy, \quad (2)$$

where R is the projection of S on the xy -plane. Hence

$$\begin{aligned} \iint_S \vec{f} \cdot \hat{n} ds &= \iint_S (f_1 \vec{i} + f_2 \vec{j} + f_3 \vec{k}) \cdot \hat{n} ds = \iint_S f_1 (\vec{i} \cdot \hat{n}) ds + \iint_S f_2 (\vec{j} \cdot \hat{n}) ds + \iint_S f_3 (\vec{k} \cdot \hat{n}) ds \\ &= \iint_S f_1 \frac{\hat{n} \cdot \vec{i}}{|\hat{n} \cdot \vec{i}|} dy dz + \iint_S f_2 \frac{\hat{n} \cdot \vec{j}}{|\hat{n} \cdot \vec{j}|} dz dx + \iint_S f_3 \frac{\hat{n} \cdot \vec{k}}{|\hat{n} \cdot \vec{k}|} dx dy \quad (\text{by (2)}) \\ &= \iint_S (f_1 dy dz + f_2 dz dx + f_3 dx dy). \end{aligned}$$

$$\iint_S (f_1 dy dz + f_2 dz dx + f_3 dx dy) = \iiint_V \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) dv.$$

Problems

Problem 9.3. Evaluate $\iint_S (x dy dz + 2y dz dx + 3z dx dy)$, where S is the closed surface of the sphere $x^2 + y^2 + z^2 = a^2$.

Solution. This is of the scalar form of the Gauss divergence theorem with $f_1 = x$, $f_2 = 2y$, $f_3 = 3z$, i.e. $\vec{f} = x\vec{i} + 2y\vec{j} + 3z\vec{k}$.

$$\operatorname{div} \vec{f} = 1 + 2 + 3 = 6.$$

$$I = \iiint_V \operatorname{div} \vec{f} dv = \iiint_V 6 dv = 6 (\text{volume of the sphere } x^2 + y^2 + z^2 = a^2) = 6 \left(\frac{4}{3} \pi a^3 \right) = 8\pi a^3.$$

Problem 9.4. If S is the surface of the sphere $x^2 + y^2 + z^2 = 1$, evaluate $\iint_S (x\vec{i} + 2y\vec{j} + 3z\vec{k}) \cdot d\vec{s}$.

Solution. $\vec{f} = x\vec{i} + 2y\vec{j} + 3z\vec{k}$, $\text{div } \vec{f} = 1 + 2 + 3 = 6$.

$$I = \iiint_V 6 \, dv = 6 (\text{volume of the sphere } x^2 + y^2 + z^2 = 1) = 6 \left(\frac{4}{3}\pi(1)^3 \right) = 8\pi.$$

Problem 9.5. Verify the divergence theorem for $\vec{f} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$, where S is the surface of the cuboid formed by the planes $x = 0$, $x = a$, $y = 0$, $y = b$, $z = 0$ and $z = c$.

Solution. **L.H.S.**

- S_1 : xy -plane, $z = 0$, $\hat{n} = -\vec{k}$: $\vec{f} \cdot \hat{n} = -z^2 = 0$, so $\iint_{S_1} = 0$.
- S_2 : parallel to the xy -plane, $z = c$, $\hat{n} = \vec{k}$: $\vec{f} \cdot \hat{n} = c^2$, $\iint_{S_2} = \int_0^a \int_0^b c^2 \, dy \, dx = abc^2$.
- S_3 : xz -plane, $y = 0$, $\hat{n} = -\vec{j}$: $\vec{f} \cdot \hat{n} = -y^2 = 0$, so $\iint_{S_3} = 0$.
- S_4 : parallel to the xz -plane, $y = b$, $\hat{n} = \vec{j}$: $\vec{f} \cdot \hat{n} = b^2$, $\iint_{S_4} = \int_0^a \int_0^c b^2 \, dz \, dx = ab^2c$.
- S_5 : yz -plane, $x = 0$, $\hat{n} = -\vec{i}$: $\vec{f} \cdot \hat{n} = 0$, so $\iint_{S_5} = 0$.
- S_6 : parallel to the yz -plane, $x = a$, $\hat{n} = \vec{i}$: $\vec{f} \cdot \hat{n} = a^2$, $\iint_{S_6} = \int_0^b \int_0^c a^2 \, dz \, dy = a^2bc$.

$$\text{L.H.S.} = abc^2 + ab^2c + a^2bc = abc(a + b + c).$$

$$\text{R.H.S.} \, \text{div } \vec{f} = 2x + 2y + 2z.$$

$$\begin{aligned} \iiint_V \text{div } \vec{f} \, dv &= \int_0^a \int_0^b \int_0^c (2x + 2y + 2z) \, dz \, dy \, dx = \int_0^a \int_0^b (2xc + 2yc + c^2) \, dy \, dx \\ &= \int_0^a (2xcb + b^2c + c^2b) \, dx = a^2cb + b^2ca + c^2ba = abc(a + b + c). \end{aligned}$$

L.H.S. = R.H.S.; hence proved.

Problem 9.6. Verify the divergence theorem for $\vec{f} = x^2\vec{i} + z\vec{j} + yz\vec{k}$ over the cube formed by $x = \pm 1$, $y = \pm 1$, $z = \pm 1$.

Solution. $\text{div } \vec{f} = 2x + 0 + y = 2x + y$.

R.H.S.

$$\begin{aligned} \iiint_V (2x + y) \, dv &= \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (2x + y) \, dz \, dy \, dx = \int_{-1}^1 \int_{-1}^1 2(2x + y) \, dy \, dx \\ &= \int_{-1}^1 [4xy + y^2]_{-1}^1 \, dx = \int_{-1}^1 8x \, dx = 4[x^2]_{-1}^1 = 0. \end{aligned}$$

L.H.S.²

- $z = 1$, $\hat{n} = \vec{k}$: $\vec{f} \cdot \hat{n} = yz = y$; $\int_{-1}^1 \int_{-1}^1 y \, dx \, dy = 0$.
- $z = -1$, $\hat{n} = -\vec{k}$: $\vec{f} \cdot \hat{n} = -yz = y$; integral = 0.
- $y = 1$, $\hat{n} = \vec{j}$: $\vec{f} \cdot \hat{n} = z$; $\int_{-1}^1 \int_{-1}^1 z \, dx \, dz = 0$.

²The face-by-face signs have been corrected here; the handwritten notes reach the same total, 0.

- $y = -1, \hat{n} = -\vec{j}$: $\vec{f} \cdot \hat{n} = -z$; integral = 0.
- $x = 1, \hat{n} = \vec{i}$: $\vec{f} \cdot \hat{n} = x^2 = 1$; $\int_{-1}^1 \int_{-1}^1 dy dz = 4$.
- $x = -1, \hat{n} = -\vec{i}$: $\vec{f} \cdot \hat{n} = -x^2 = -1$; integral = -4.

L.H.S. = $0 + 0 + 0 + 0 + 4 - 4 = 0 =$ R.H.S. Hence the divergence theorem is verified.